

## 2. Root finding

### 2.1. Introduction to Root Finding

The root-finding problem is a fundamental task in numerical analysis, where we aim to find solutions  $x^*$  to the equation:

$$f(x^*) = 0 \tag{2.1}$$

for a given continuous function  $f(x)$ . The solutions to this equation,  $x^*$ , are called the *roots* or *zeros* of the function. In many practical applications, these roots may not be easily solvable analytically, necessitating the use of numerical methods to approximate them. The root finding problem can easily be extended to solve for a non-zero value  $y_0$  by reformulating the problem

$$f(x^*) = y_0 \tag{2.2}$$

to the problem

$$g(x^*) = f(x^*) - y_0 = 0 \tag{2.3}$$

and solve the equation  $g(x^*) = 0$ . This ansatz can also be extended to more complex functions. Say, you would like to find  $x^*$  for which  $f(x^*) = h(x^*)$ , then define  $g(x) = f(x) - h(x)$  and solve for  $g(x^*) = 0$ .

Let us look at two simple examples

- Quadratic Equation:  $f(x) = x^2 - 4 = 0$ . The exact roots are  $x = \pm 2$ .
- Consider a more complex equation:  $f(x) = \sin(x) - \frac{x}{2} = 0$ . There is no simple closed-form solution, and numerical methods are required to approximate the roots.

It is typically a good start to *look* at the function and inspect the approximate range of values before starting to find roots numerically.

### 2.2. Bisection Method

The bisection method is a simple and robust numerical method to find the root of a continuous function within a given interval. The method works by repeatedly halving the interval and selecting the subinterval in which the root must lie, based on the Intermediate Value Theorem.

## 2. Root finding

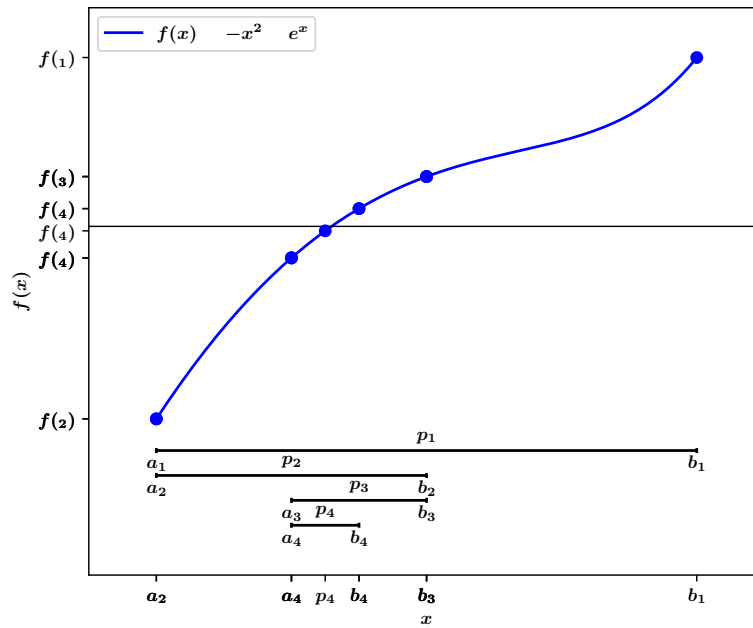


Figure 2.1.: Illustration of the bisection method.

### 2.2.1. Description of the Algorithm

Let  $f(x)$  be a continuous function, and let  $a$  and  $b$  be two points such that  $f(a)f(b) < 0$ , meaning that the function changes sign over the interval  $[a, b]$ . This implies, by the Intermediate Value Theorem, that there is at least one root in this interval. **Note:** If the function is not continuous but bounded, then  $f$  will have a step. If  $f$  is not bounded then the function can have a pole. The bisection method proceeds as follows:

1. begin by setting  $a_1 := a, b_1 := b$
2. Compute the midpoint  $p := \frac{a+b}{2}$ .
3. Evaluate  $f(p)$ .
4. If  $f(p) = 0$ , then  $p$  is the root.
5. If  $f(a)f(p) < 0$ , then the function changed sign in the interval  $[a, p]$ , i.e. the root must be there, so set  $b_2 = p$ .
6. If  $f(p)f(b) < 0$ , the root lies in the interval  $[p, b]$ , so set  $a_2 = p$ .
7. Repeat the process until the interval  $[a_n, b_n]$  is sufficiently small.

We would like to iterate a bit more on possible stopping criteria:

- the most common criterion is a check if  $(b_n - a_n) < \text{TOL}$ , i.e. a check on the total size of the interval. This fails for step functions and poles. The tricky part is that the bisection methods stops and will give an answer, which is wrong.
- Alternatively, we can check for convergence of subsequent steps:  $|p_n - p_{n-1}| < \epsilon$ .
- A third possibility is the check for the function value  $|f(p_n)| < \epsilon$ . This is more stringent and would never be fulfilled for a pole, so the method could detect the non-convergence.

In practice it is recommended to always limit the number of steps that an algorithm is allowed to take before it has to stop. This will catch a wrong behaviour in cases such as

- a function does not converge
- $a_n = b_n$  within the numerical precision
- some NaNs or other weird configurations are present.

### 2.2.2. Example: Finding the Root of $f(x) = x^2 - 2$

Let  $f(x) = x^2 - 2$ . We want to find  $\sqrt{2}$ , which is the root of  $f(x)$ . Choose  $a = 1$  and  $b = 2$ , where  $f(1) = -1$  and  $f(2) = 2$ . Applying the bisection method:

1. Compute the midpoint  $m = 1.5$ , and  $f(1.5) = 0.25$ .
2. Since  $f(1)f(1.5) = -1 \cdot 0.25 < 0$ , the root lies in  $[1, 1.5]$ .
3. Next, compute  $m = 1.25$ , and  $f(1.25) = -0.4375$ .
4. Continue this process, halving the interval until the desired precision is reached.

This procedure produces the following output:

| Iteration | Lower   | Upper   | Middle  | Error        |
|-----------|---------|---------|---------|--------------|
| 1         | 1.00000 | 2.00000 | 1.50000 | 0.5000000000 |
| 2         | 1.00000 | 1.50000 | 1.25000 | 0.2500000000 |
| 3         | 1.25000 | 1.50000 | 1.37500 | 0.1250000000 |
| 4         | 1.37500 | 1.50000 | 1.43750 | 0.0625000000 |
| 5         | 1.37500 | 1.43750 | 1.40625 | 0.0312500000 |
| 6         | 1.40625 | 1.43750 | 1.42188 | 0.0156250000 |
| 7         | 1.40625 | 1.42188 | 1.41406 | 0.0078125000 |
| 8         | 1.41406 | 1.42188 | 1.41797 | 0.0039062500 |
| 9         | 1.41406 | 1.41797 | 1.41602 | 0.0019531250 |
| 10        | 1.41406 | 1.41602 | 1.41504 | 0.0009765625 |
| 11        | 1.41406 | 1.41504 | 1.41455 | 0.0004882812 |
| 12        | 1.41406 | 1.41455 | 1.41431 | 0.0002441406 |
| 13        | 1.41406 | 1.41431 | 1.41418 | 0.0001220703 |
| 14        | 1.41418 | 1.41431 | 1.41425 | 0.0000610352 |
| 15        | 1.41418 | 1.41425 | 1.41422 | 0.0000305176 |
| 16        | 1.41418 | 1.41422 | 1.41420 | 0.0000152588 |
| 17        | 1.41420 | 1.41422 | 1.41421 | 0.0000076294 |

## 2. Root finding

### 2.2.3. Advantages and Disadvantages

#### Advantages:

- Simple and robust, guaranteed to converge if  $f(x)$  is continuous and changes sign over the interval.
- Does not require derivatives.

#### Disadvantages:

- Converges slowly compared to other methods.
- Requires that the function change sign over the interval.

## 2.3. Fixed-Point Iteration

### 2.3.1. Preliminaries

- A number  $p$  is a **fixed point** for a function  $g$  if  $g(p) = p$ .
- Can transform a function to a fixed-point problem, for example

$$\begin{aligned}g(x) &= x - f(x) \Leftrightarrow f(x) = x - g(x) \\ f(p) &= p - g(p) = 0,\end{aligned}$$

so  $f(x)$  has a root at  $p$ .

#### Algorithm:

Start with initial guess  $p_0$ :

1.  $p = g(p_0)$
2. If  $|p - p_0| \leq \text{TOL}$  **then** done
3. else:  $p_0 = p$  (update  $p_0$ )

#### Illustration:

- Sketches typically show the zig-zag or spiral convergence between  $y = g(x)$  and  $y = x$  as illustrated in Fig. 2.2.

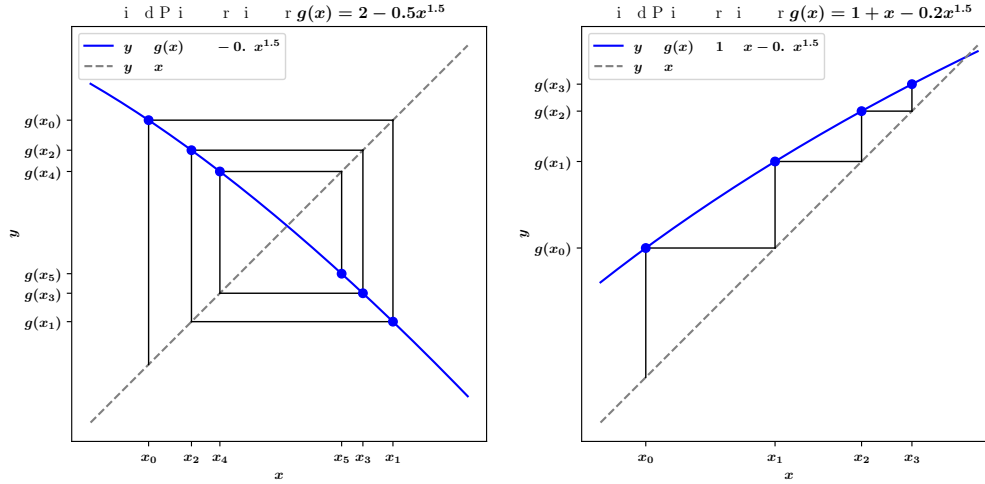


Figure 2.2.: Illustration of the fixed-point iteration.

### Fixed-Point Theorem and Examples

Let us consider the following example for a function

$$x^3 + 4x^2 - 10 = 0,$$

which has a unique root in  $[1, 2]$ . There are now many ways to transform this function into a fixed-point problem. A few possible forms are

1.  $4x^2 = 10 - x^3 \Rightarrow x^2 = \frac{1}{4}(10 - x^3) \Rightarrow x = \pm \frac{1}{2}(10 - x^3)^{1/2}$
2.  $x^2(x + 4) = 10 \Rightarrow x = \left(\frac{10}{x+4}\right)^{1/2}$
3.  $x = x - 0 = x - \frac{x^3 + 4x^2 - 10}{3x^2 + 8x}$

### Fixed-Point Theorem:

Let  $g \in C[a, b]$  such that  $g(x) \in [a, b]$  for all  $x \in [a, b]$ . In addition, if  $g$  is differentiable on  $(a, b)$  and a constant  $0 < k < 1$  exists such that:

$$|g'(x)| \leq k \quad \forall x \in (a, b),$$

then for any number  $p_0 \in [a, b]$ , the sequence:

$$p_n = g(p_{n-1}), \quad n \geq 1$$

converges to the **unique fixed point** in  $[a, b]$ .

## 2. Root finding

### Corollary:

See your favourite math book. In words this means: an interval  $[a, b]$  needs to be mapped into itself.

**Idea:** Manipulate root-finding problem into appropriate fixed-point problem.

## 2.4. Newton-Raphson Method

The Newton-Raphson method is a more efficient algorithm for root finding that uses both the function and its derivative. It converges faster than the bisection method, but requires the function to be differentiable and may fail if the derivative is zero or the initial guess is poor.

### 2.4.1. Preliminaries

- Assume  $f \in C^2[a, b]$ .
- Let  $p_0 \in [a, b]$  be an approximation to  $p$  such that  $f(p_0) \neq 0$  and  $|p - p_0|$  is “small”.

We can formulate the function  $f$  as a Taylor polynomial

$$f(p) = f(p_0) + (p - p_0)f'(p_0) + \frac{(p - p_0)^2}{2}f''(\xi(p))$$

where  $\xi(p)$  lies between  $p$  and  $p_0$ . If  $f(p) = 0$  (i.e.,  $p$  is the root), then we find:

$$\begin{aligned} 0 &\approx f(p_0) + (p - p_0)f'(p_0) \\ p &\approx p_0 - \frac{f(p_0)}{f'(p_0)} =: p_1 \end{aligned}$$

### 2.4.2. Algorithm:

Assuming an initial guess / approximation for the root,  $p_0$ , we can simply define the algorithm as a sequence:

$$p_n = p_{n-1} - \frac{f(p_{n-1})}{f'(p_{n-1})}, \quad n \geq 1$$

An illustration of the sequence following the first derivatives, i.e. the tangents along the function, is shown in Fig. 2.3. Again, suitable stopping criteria need to be applied:

- $|p_n - p_{n-1}| < \epsilon$  (absolute)
- $\left| \frac{p_n - p_{n-1}}{p_n} \right| < \epsilon$  (relative)
- $|f(p_n)| < \epsilon$  (function value small)

**Note:** None of the above give precise error estimates, only upper limits!

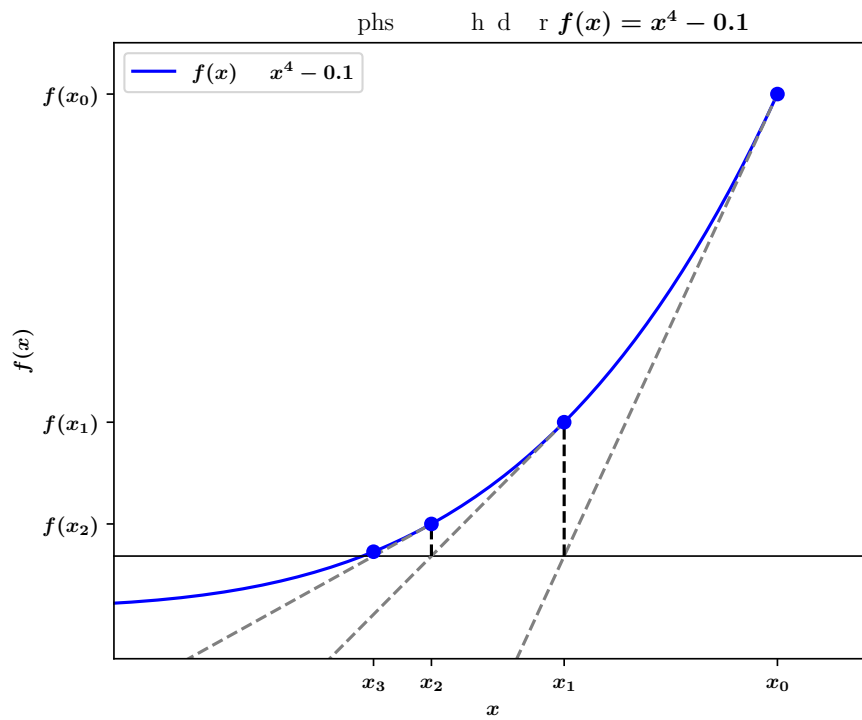


Figure 2.3.: Illustration of the Newton-Raphson method. Note that here the example function is different from the one in the text to better highlight the method.

## 2. Root finding

### Fixed-Point example revisited:

We note that we can map Newton's Method into a Fixed-Point Iteration

$$p_n = g(p_{n-1}) = p_{n-1} - \frac{f(p_{n-1})}{f'(p_{n-1})}, \quad n \geq 1$$

with the function and  $f(x)$  and its first derivative  $f'(x)$ :

$$f(x) = x^3 + 4x^2 - 10$$

$$f'(x) = 3x^2 + 8x$$

$$g(x) = x - \frac{x^3 + 4x^2 - 10}{3x^2 + 8x}$$

### Newton's Method as Fixed-Point Iteration:

$$p_n = g(p_{n-1}) = p_{n-1} - \frac{f(p_{n-1})}{f'(p_{n-1})}, \quad n \geq 1$$

We use again the same example as in the bisection method ( $f(x) = x^2 - 2$ ) and use  $x_0 = 2.5$  as a starting point. In the iterations we find

| Iteration | x            | f(x)         | Error        |
|-----------|--------------|--------------|--------------|
| 1         | 2.5000000000 | 4.2500000000 | 0.8500000000 |
| 2         | 1.6500000000 | 0.7225000000 | 0.2189393939 |
| 3         | 1.4310606061 | 0.0479344582 | 0.0167478785 |
| 4         | 1.4143127276 | 0.0002804914 | 0.0000991617 |
| 5         | 1.4142135658 | 0.0000000098 | 0.0000000035 |

### 2.4.3. Advantages and Disadvantages

#### Advantages:

- Fast convergence when near the root, often quadratic convergence.
- Efficient for well-behaved functions.

#### Disadvantages:

- Requires the computation of the derivative  $f'(x)$ .
- Can fail if  $f'(x)$  is zero or if the initial guess is too far from the root.
- Not guaranteed to converge, especially for poorly chosen starting points.

## 2.5. Secant Method

The Secant method is an iterative numerical technique used to find the roots of a function  $f(x) = 0$ . Unlike the Newton-Raphson method, the Secant method does not require knowledge of the derivative  $f'(x)$ , making it useful for functions where derivatives are difficult or expensive to compute.

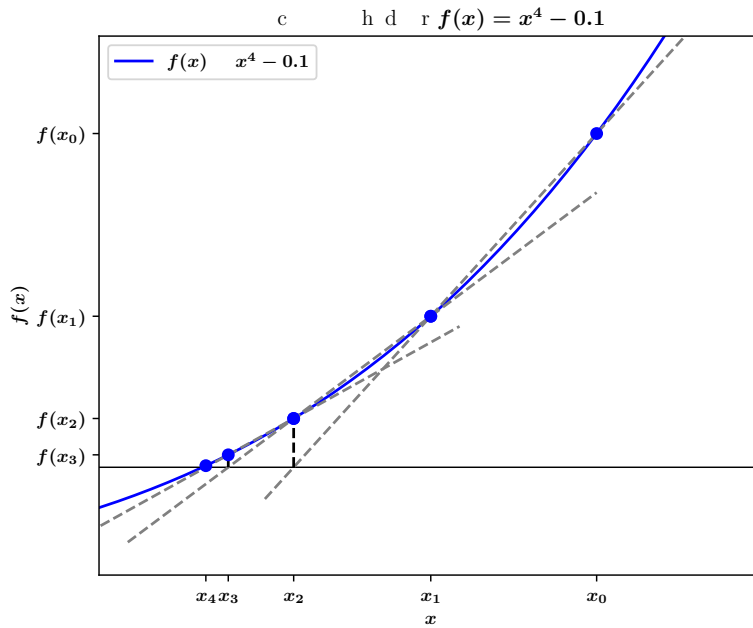


Figure 2.4.: Illustration of the Secant method.

### 2.5.1. Description of the algorithm

The Secant method requires two initial guesses,  $x_0$  and  $x_1$ , which should be close to the root. The method then approximates the derivative by using a secant line passing through these two points. The iterative formula for the Secant method is:

$$x_{n+1} = x_n - f(x_n) \cdot \frac{x_n - x_{n-1}}{f(x_n) - f(x_{n-1})}. \quad (2.4)$$

An illustration of the method is shown in Fig. 2.4. This formula is repeated until convergence is reached, typically when the difference between successive approximations is below a specified tolerance level, or a maximum number of iterations is reached.

### 2.5.2. Example

Let's consider finding the root of the function  $f(x) = x^2 - 4$ , which has a known root at  $x = 2$ . We'll use initial guesses  $x_0 = 1$  and  $x_1 = 3$ .

- **Iteration 1:**

$$x_2 = x_1 - f(x_1) \cdot \frac{x_1 - x_0}{f(x_1) - f(x_0)} \quad (2.5)$$

where  $f(x_1) = 5$  and  $f(x_0) = -3$ .

- **Continue Iterating:** Continue using the same formula with updated values of  $x$  until  $f(x_n)$  is close to zero.

## 2. Root finding

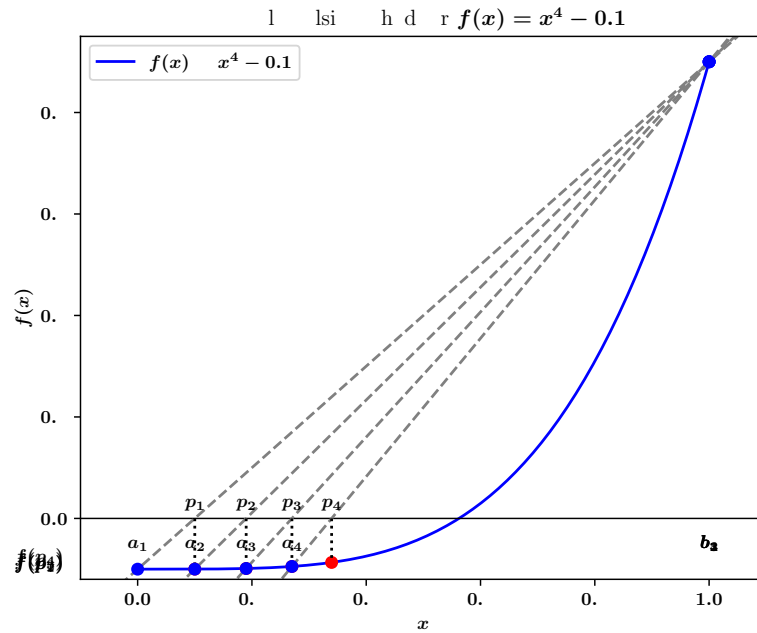


Figure 2.5.: Illustration of the Regula Falsi method.

### 2.5.3. Advantages and Disadvantages

Advantages:

- **No Need for Derivative:** Unlike the Newton-Raphson method, the Secant method does not require knowledge of  $f'(x)$ .
- **Faster Convergence:** Generally converges faster than the Bisection method, especially if initial guesses are close to the root.

Disdvantages:

- **Initial Guesses are Crucial:** Poor initial guesses can lead to slow convergence or divergence.
- **Not Guaranteed to Converge:** Unlike the Bisection method, the Secant method does not guarantee convergence.

## 2.6. False Position (Regula Falsi) Method

The False Position method, also known as the Regula Falsi method, is an iterative root-finding technique that combines elements of the Bisection and Secant methods. It is particularly useful for functions where a root is bracketed within an interval  $[a, b]$  such that  $f(a)$  and  $f(b)$  have opposite signs, ensuring at least one root exists in the interval.

### 2.6.1. Description of the algorithm

Starting with two initial guesses,  $a$  and  $b$ , where  $f(a)$  and  $f(b)$  have opposite signs, the False Position method uses a linear interpolation to approximate the root. The formula for the next estimate  $c$  is:

$$c = b - f(b) \cdot \frac{b - a}{f(b) - f(a)} \quad (2.6)$$

After calculating  $c$ , the method checks the sign of  $f(c)$ :

- If  $f(c) \cdot f(a) < 0$ , then the root lies between  $a$  and  $c$ , so we set  $b = c$ .
- If  $f(c) \cdot f(b) < 0$ , then the root lies between  $c$  and  $b$ , so we set  $a = c$ .

An illustration is shown in Fig. 2.5. This process is repeated until the function value at  $c$  is close to zero (within a specified tolerance), or until a maximum number of iterations is reached.

### 2.6.2. Example

Consider finding the root of the function:

$$f(x) = x^3 - x - 2 \quad (2.7)$$

with initial guesses  $a = 1$  and  $b = 2$ , where  $f(a) = -2$  and  $f(b) = 2$ .

1. Iteration 1:

$$c = b - f(b) \cdot \frac{b - a}{f(b) - f(a)} = 2 - 2 \cdot \frac{2 - 1}{2 - (-2)} = 1.5 \quad (2.8)$$

Calculate  $f(c) = -0.125$ .

2. Subsequent Iterations:

Continue using the same formula with updated values of  $a$  or  $b$  depending on the sign of  $f(c)$ , until convergence is achieved.

### 2.6.3. Advantages and Disadvantages

**Advantages:**

- **Guaranteed Convergence:** As long as the root is bracketed initially, the method is guaranteed to converge.
- **No Derivative Required:** Like the Bisection method, the False Position method does not require derivative information.

**Disdvantages:**

- **Slower Convergence for Certain Functions:** If  $f(x)$  is steep near one of the interval's ends, the method may converge slowly as it tends to favor the steeper side, see Fig. 2.5.

## 2. Root finding

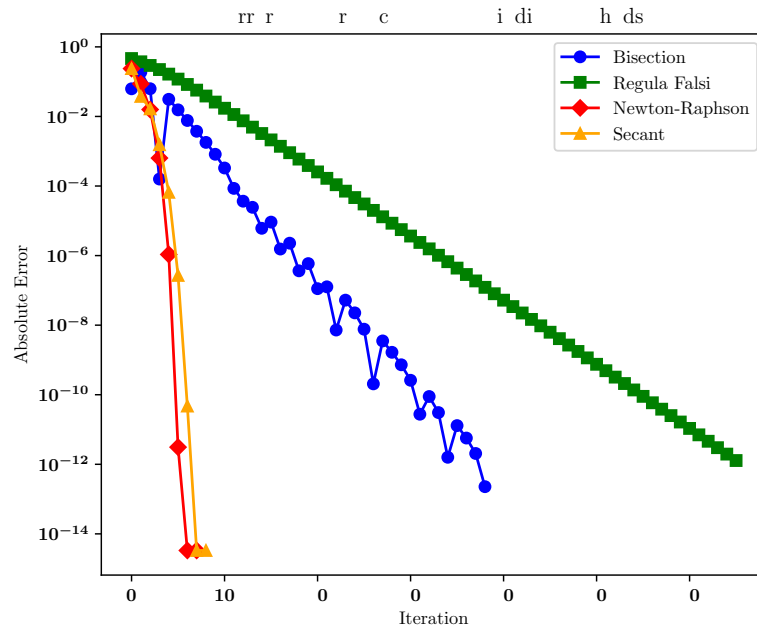


Figure 2.6.: Comparison of the discussed methods. Error convergence as a function of iteration

## 2.7. Comparison of Root-Finding Methods

We compare four root-finding methods — Bisection, Regula Falsi, Newton-Raphson, and Secant — applied to the nonlinear function  $f(x) = x^4 - 0.1$  on the interval  $[0, 1]$ . Bracketing methods (Bisection and Regula Falsi) use the initial interval  $[0, 1]$ , while Newton-Raphson and Secant start from the initial guesses  $x_0 = 0.8$  and  $x_1 = 0.6$ . The figure shows the convergence behavior of each method in terms of the absolute error as a function of iteration count on a log-log scale. As expected, Bisection and Regula Falsi converge linearly. Newton-Raphson demonstrates quadratic convergence near the root, whereas the Secant method shows superlinear convergence, performing faster than bracketing methods but slightly slower than Newton-Raphson. This comparison highlights the trade-off between robustness and speed of convergence in classical root-finding algorithms.

## 2.8. Multiple Roots

So far we have discussed the problem of  $f(x) = 0$  in the case that  $f'(x) \neq 0$ , i.e. the function actually passes through the  $y = 0$  line with a non-zero slope. This is however not the case for multiple roots. The function  $f(x) = x^2$  only touches the  $y = 0$  line with a slope of  $f'(0) = 0$ . The odd polynomial  $f(x) = x^3$  crosses the  $y = 0$  line, but still has a vanishing slope at the root ( $f'(0) = 0$ ).

The Bisection method is designed to only work with function values of opposite sign for the interval boundaries  $a_n$  and  $b_n$ . Similarly, the Secant method will not work for the same reason. The Newton-Raphson method does not put any restriction on the absolute function values, but needs the slope to be non-zero at the root,

$$p_n = p_{n-1} \frac{f(p_{n-1})}{f'(p_{n-1})}. \quad (2.9)$$

The convergence will fail to find the exact value by definition and depending on step size and accuracy of the computation will lead to unwanted behaviour (NaN, large values).

### 2.8.1. Definition Multiplicity of a root

A solution  $p$  of  $f(x) = 0$  is a **zero of multiplicity**  $m$  of  $f$  if for  $x \neq p$ , we can write

$$f(x) = (x - p)^m q(x), \quad \text{where } \lim_{x \rightarrow p} q(x) \neq 0. \quad (2.10)$$

In words, this means that the root  $p$  needs to factor out to power  $m$ ,  $(x - p)^m$ , and  $q(x)$  represents the part of  $f$  that does not contribute to the root, i.e. does not approach zero at  $p$ .

To make use of that multiplicity in practice, we need a method to identify simple roots for a function, which is given by the following theorem:

**Theorem** The function  $f \in C^1[a, b]$  has a simple zero at  $p$  in  $(a, b)$  if and only if  $f(p) = 0$  but  $f'(p) \neq 0$ .

**Proof** We need to proof both directions: (a) if  $f$  has a simple root then  $f(p) = 0$  and  $f'(p) \neq 0$  and (b) if  $f(p) = 0$  and  $f'(p) \neq 0$  then  $f(p)$  has a root with multiplicity  $m = 1$ .

- (a) If  $f$  has a simple zero at  $p$ , then  $f(p) = 0$  and  $f(x) = (x - p)q(x)$ , where  $\lim_{x \rightarrow p} q(x) \neq 0$ . Since  $f \in C^1[a, b]$ ,

$$f'(p) = \lim_{x \rightarrow p} f'(x) = \lim_{x \rightarrow p} [q(x) + (x - p)q'(x)] = \lim_{x \rightarrow p} q(x) \neq 0. \quad (2.11)$$

- (b) Conversely, if  $f(p) = 0$  and  $f'(p) \neq 0$ , expand  $f$  in a zeroth Taylor polynomial about  $p$ . Then

$$f(x) = f(p) + f'(\xi(x))(x - p) = (x - p)f'(\xi(x)), \quad (2.12)$$

where  $\xi(x)$  is between  $x$  and  $p$ . Since  $f \in C^1[a, b]$ ,

$$\lim_{x \rightarrow p} f'(\xi(x)) = f' \left( \lim_{x \rightarrow p} \xi(x) \right) = f'(p) \neq 0. \quad (2.13)$$

Letting  $q = f' \circ \xi$  gives  $f(x) = (x - p)q(x) = (x - p)f'(\xi(x))$ , where  $\lim_{x \rightarrow p} q(x) \neq 0$ . Thus,  $f$  has a simple zero at  $p$ .

## 2. Root finding

**Theorem** The function  $f \in C^m[a, b]$  has a zero of multiplicity  $m$  at  $p$  in  $(a, b)$  if and only if

$$0 = f(p) = f'(p) = \dots = f^{(m-1)}(p) \quad \text{but} \quad f^{(m)}(p) \neq 0 \quad (2.14)$$

**Example** Let us examine the example  $f(x) = e^x - x - 1$ . The function and first two derivatives are given as

$$\begin{aligned} f(x) &= e^x - x - 1 & f'(x) &= e^x - 1 & f''(x) &= e^x \\ f(0) &= 0 & f'(0) &= 0 & f''(0) &= 1 \end{aligned}$$

The function thus has a zero of multiplicity 2 at  $p = 0$ .

### 2.8.2. Solution for multiple roots

One method of handling the problem of multiple roots of a function  $f$  is to define

$$\mu(x) = \frac{f(x)}{f'(x)}. \quad (2.15)$$

If  $p$  is a zero of  $f$  of multiplicity  $m$  with  $f(x) = (x - p)^m q(x)$ , then

$$\mu(x) = \frac{f(x)}{f'(x)} = \frac{(x - p)^m q(x)}{m(x - p)^{m-1} q(x) + (x - p)^m q'(x)} \quad (2.16)$$

$$= (x - p) \frac{q(x)}{mq(x) + (x - p)q'(x)}. \quad (2.17)$$

Note that we can define this ratio  $\mu(x)$  since we divide here by the function  $f'(x)$  for general  $x$ , not  $f'(p)$ , which is 0. We have factored out  $(x - p)$ , so we see that  $\mu(x)$  has a zero at  $p$ . To make use of  $\mu$  conveniently, we need to show that it has a zero of multiplicity 1 at  $p$ . Based on the definition of the multiplicity, we need to show that the second term with the fraction  $\neq 0$ . We know that  $q(p) \neq 0$  by definition, so

$$\frac{q(p)}{mq(p) + (p - p)q'(p)} = \frac{1}{m} \neq 0. \quad (2.18)$$

This means that  $p$  is a simple zero of  $\mu(x)$ , and – based on the theorem above – therefore has non-vanishing first derivative. This means that Newton's method can then be applied to  $\mu(x)$  to give

$$g(x) = x - \frac{\mu(x)}{\mu'(x)} \quad (2.19)$$

$$= x - \frac{f(x)/f'(x)}{\{[f'(x)]^2 - [f(x)]f''(x)\}/[f'(x)]^2}, \quad (2.20)$$

which simplifies to

$$g(x) = x - \frac{f(x)f'(x)}{[f'(x)]^2 - f(x)f''(x)}. \quad (2.21)$$

If  $g$  has the required continuity conditions, functional iteration applied to  $g$  will be quadratically convergent regardless of the multiplicity of the zero of  $f$ . Theoretically, the only drawback to this method is the additional calculation of  $f''(x)$  and the more laborious procedure of calculating the iterates. In practice, however, multiple roots can cause serious round-off problems because the denominator of Eq. (2.21) consists of the difference of two numbers both of which are close to 0.

## 2.9. Some further methods and combinations

Before we also mention two advanced methods, let us revisit the types of the methods used before.

- bisection: simple bracketing
- Newton/secant: iterative Taylor approximation
- Regula falsi: interpolation

The idea is now to combine these methods and invent a fast *and* stable method.

- Müller's Method

Uses quadratic interpolation on three points to approximate the root. Converges faster than methods based on linear interpolation (like the secant method) but may introduce complex roots if the function has no real roots in the interval.

- Brent's Method

A hybrid method combining bisection, secant, and inverse quadratic interpolation. Known for being robust and efficient, it doesn't require derivative information and combines the speed of the secant and interpolation methods with the reliability of the bisection method.

## 2.10. Conclusion

The bisection method provides guaranteed convergence for a value  $x$ , but will fail at a pole and the convergence is slower, while the Newton-Raphson method is faster but may not always converge. If the derivatives of the function are not known, one can approximate it using the differences, which leads to the Secant method.



# A. Physics

## A.1. Response Functions: Heat Capacity and Susceptibility

### A.1.1. What Are Response Functions?

In thermodynamics and statistical physics, a **response function** tells us how a macroscopic observable reacts to a small change in a control parameter:

- The **heat capacity**  $C_V$  measures how strongly the energy responds to a change in temperature:

$$C_V = \left( \frac{\partial \langle E \rangle}{\partial T} \right)_V$$

*Illustratively: How much does the energy fluctuate as we heat the system up?*

- The **magnetic susceptibility**  $\chi$  measures how strongly the magnetisation responds to a magnetic field:

$$\chi = \left( \frac{\partial \langle M \rangle}{\partial h} \right)_T$$

*Illustratively: How easily can we magnetise the system with a weak field?*

Surprisingly, these quantities can be computed *without changing*  $T$  or  $h$ , by observing the *natural fluctuations* in equilibrium.

### A.1.2. Canonical Ensemble

Let the system be in contact with a heat bath at fixed temperature  $T$ . The probability of microstate  $i$  is:

$$P_i = \frac{e^{-\beta E_i}}{Z}, \quad \beta = \frac{1}{k_B T}, \quad Z = \sum_i e^{-\beta E_i}$$

### A.1.3. Heat Capacity from Energy Fluctuations

The average energy is:

$$\langle E \rangle = \sum_i E_i P_i = \frac{1}{Z} \sum_i E_i e^{-\beta E_i}$$

## A. Physics

Differentiate with respect to  $T$ , using  $\frac{d}{dT} = -\frac{1}{k_B T^2} \frac{d}{d\beta}$ :

$$C_V = \frac{d\langle E \rangle}{dT} = -\frac{1}{k_B T^2} \cdot \frac{d\langle E \rangle}{d\beta}$$

Now compute:

$$\frac{d\langle E \rangle}{d\beta} = \frac{d}{d\beta} \left( \frac{1}{Z} \sum_i E_i e^{-\beta E_i} \right) = -\langle E^2 \rangle + \langle E \rangle^2 = -\text{Var}(E)$$

So:

$$C_V = \frac{1}{k_B T^2} (\langle E^2 \rangle - \langle E \rangle^2) = k_B \beta^2 (\langle E^2 \rangle - \langle E \rangle^2)$$

### A.1.4. Magnetic Susceptibility from Magnetisation Fluctuations

Introduce a magnetic field  $h$  into the Hamiltonian:

$$\mathcal{H} = \mathcal{H}_0 - hM$$

Then the probability becomes:

$$P_i = \frac{e^{-\beta(E_i - hM_i)}}{Z}$$

The average magnetisation is:

$$\langle M \rangle = \sum_i M_i P_i = \frac{1}{Z} \sum_i M_i e^{-\beta(E_i - hM_i)} = \frac{\partial}{\partial(\beta h)} \ln Z$$

Differentiate with respect to  $h$ :

$$\chi = \frac{\partial \langle M \rangle}{\partial h} = \frac{\partial^2 \ln Z}{\partial(\beta h)^2} \cdot \left( \frac{\partial(\beta h)}{\partial h} \right)^2 = \beta (\langle M^2 \rangle - \langle M \rangle^2)$$

Hence:

$$\chi = \frac{1}{k_B T} (\langle M^2 \rangle - \langle M \rangle^2) = \beta (\langle M^2 \rangle - \langle M \rangle^2)$$

This expression is rigorous in the canonical ensemble. In computational models where  $k_B = 1$ , it simplifies to  $\chi = \beta \text{Var}(M)$ .

### A.1.5. Interpretation

- The response functions  $C_V$  and  $\chi$  are directly related to *fluctuations* of their conjugate variables (energy and magnetisation).
- These are exact results for equilibrium systems in the canonical ensemble.
- Near a phase transition, these fluctuations often diverge in the thermodynamic limit, making  $C_V$  and  $\chi$  useful diagnostics.